

NLP Assignment 3

Implementing BERT for Fake News Classification

Harsh Jain, Dishant Vishwakarma
25CS60R38, 25CS60R36

1 Introduction

In this assignment, we implement a BERT-based model for fake news classification using the Fakedit dataset. The assignment consists of four tasks: implementing BERT from scratch, preprocessing the dataset, fine-tuning a pretrained BERT model, and interpreting predictions using Integrated Gradients.

2 Task 1: BERT Implementation from Scratch

We implemented a simplified BERT encoder consisting of:

- Token Embeddings
- Positional Embeddings
- Token-Type Embeddings
- Multi-Head Self Attention
- Feed Forward Network
- Transformer Encoder Layers
- Classification Head

2.1 Embedding Shapes

The model was tested on a sample sentence and the following shapes were observed:

- Token Embeddings: $(1, 20, 512)$
- Positional Embeddings: $(1, 20, 512)$
- Encoder Output: $(1, 20, 512)$
- Classifier Output: $(1, 2)$

2.2 Model Parameter Shapes

- Token Embedding: (30522, 512)
- Positional Embedding: (128, 512)
- Token-Type Embedding: (2, 512)
- Feed Forward Layer: (2048, 512)
- Classifier: (2, 512)

3 Task 2: Data Preprocessing

3.1 Dataset Construction

The dataset was created by combining:

- all_train.tsv
- all_test_public.tsv
- all_validate.tsv

The following columns were retained:

- clean_title
- 2_way_label
- id

3.2 Dataset Statistics

- Original dataset size: 253,498

3.3 Balanced Dataset

- Fake posts: 2500
- Real posts: 2500

3.4 Train-Test Split

Dataset	Real (0)	Fake (1)	Total
Train	2000	2000	4000
Test	500	500	1000

4 Task 3: Fine-Tuning BERT

4.1 Model

We used the pretrained `bert-base-uncased` model with a classification head.

4.2 Hyperparameters

Parameter	Value
Epochs	2
Batch Size	8
Learning Rate	2e-5
Max Sequence Length	128
Loss Function	Cross Entropy

4.3 Training

The model was fine-tuned using cross-entropy loss. Input sequences were truncated to a maximum length of 128 tokens to balance computational efficiency and performance.

4.4 Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score

4.5 Results

Metric	Value
Accuracy	79.6%
Precision	0.7587
Recall	0.8680
F1-score	0.8097

4.6 Confusion Matrix

$$\begin{bmatrix} 362 & 138 \\ 66 & 434 \end{bmatrix}$$

5 Task 4: Integrated Gradients

5.1 Method

Integrated Gradients (IG) was used to interpret the predictions made by the fine-tuned BERT model by assigning importance scores to input tokens. The method computes the contribution of each token by integrating gradients along a path from a baseline to the actual input.

5.2 Setup

- Attribution Method: Layer Integrated Gradients
- Layer Used: Embedding Layer

- Baseline: Zero embeddings
- Number of Steps: 100

5.3 Procedure

- Compute embeddings for input tokens
- Use zero embeddings as baseline
- Compute attributions using IG
- Aggregate attribution scores across embedding dimension
- Map token IDs back to tokens
- Remove special tokens ([CLS], [SEP], [PAD])
- Extract top-5 important tokens

5.4 Examples and Token Attributions

Example 1

- Text: santa uses sign language with deaf boy youtube
- True Label: Fake (1), Predicted Label: Fake (1)
- boy (0.2753), uses (0.2184), deaf (0.1797), sign (0.1315), language (0.1290)

Example 2

- Text: honeywell workers say lockout aims to destroy union its corporate greed
- True Label: Fake (1), Predicted Label: Fake (1)
- workers (0.5497), destroy (-0.3426), say (0.3111), aims (0.2211), corporate (0.2187)

Example 3

- Text: this dog on a raft
- True Label: Fake (1), Predicted Label: Fake (1)
- dog (0.7456), this (0.2946), raft (0.0696), on (0.0371), a (-0.0018)

Example 4

- Text: a koala hiding in a tree
- True Label: Real (0), Predicted Label: Fake (1)
- ala (-0.5405), in (0.4366), a (0.4047), hiding (0.2110)

Example 5

- Text: a woodpecker put holes in my wood fence

- True Label: Real (0), Predicted Label: Fake (1)
- pe (-0.1661), in (0.1096), wood (-0.1077), fence (-0.1019), holes (-0.0890)

Example 6

- Text: pga championship cancelled after disastrous publicity stunt the yew norker
- True Label: Real (0), Predicted Label: Real (0)
- the (0.5900), disastrous (0.4556), publicity (0.2590), nor (0.2497)

5.5 Observations

- Important tokens correspond to semantically meaningful words such as *workers*, *dog*, and *disastrous*.
- Both positive and negative attribution values indicate the direction of influence on predictions.
- Subword tokens (e.g., *ala*, *pe*) appear due to BERT tokenization.
- The model performs well on clear fake news examples but struggles with short or ambiguous real-world sentences.
- Misclassifications (e.g., koala sentence and woodpecker sentence) highlight limitations of the model in handling simple descriptive text.

5.6 Convergence Delta

The convergence delta values vary across samples. Smaller values indicate stable attribution estimates, while larger values (e.g., in shorter or ambiguous sentences) suggest approximation errors. Increasing the number of integration steps can further improve convergence.

6 Conclusion

The assignment successfully demonstrates the implementation of BERT from scratch, fine-tuning on a classification task, and interpreting predictions using Integrated Gradients. The model achieved good performance and the interpretability analysis provided insights into model decision-making.